

Roller-Coaster Geometry

Framed curves, rider observations, and regular rail offsets

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Status. This document supersedes the bilingual working draft v1.0. It is migrated to GO Core v0.2 and the Frames-Forces-Dissipation Interface v0.1. It is a differential-geometric and observation laboratory, not a track design, vehicle model, human-tolerance model, or safety certificate.

Abstract

A roller-coaster centerline does not encode vehicle orientation, rail-contact paths, or rider measurements. We define a track as a unit-speed curve $\gamma(s)$, a nonsingular adapted frame $R(s) \in \text{SO}(3)$, and either a prescribed speed profile or a declared dynamical law. With $R^T R' = [\kappa_R]_\times$, the vehicle angular velocity is $\Omega^B = v\kappa_R$. The inertial acceleration is

$$a^I = v \frac{dv}{ds} T + v^2 T',$$

while an ideal vehicle accelerometer reads the body-frame specific force

$$f^B = R^T(a^I - g^I).$$

The dimensionless load vector $G = f^B/g_0$ and its scalar norm $G = \|G\|$ are separate quantities. Body-frame jerk includes the frame-rotation correction and is not merely $R^T \dot{a}$. Ideal rail offsets $\gamma_\pm = \gamma \pm (d/2)e_2$ are regular under the sufficient bound $(d/2) \sup \|e'_2\| < 1$; global nonintersection additionally requires a tubular-neighborhood hypothesis. The labeled length split $\Delta L = L_+ - L_-$ changes sign under rail exchange, whereas $|\Delta L|$ is exchange invariant. Dissipation and moving support work follow the common energy contract. All claims are qualified by their transformation group and model status.

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1 Scope and protocol

An observation protocol λ specifies the world frame, adapted-frame convention, clock, arc-length chart, sensor axes, sampling, calibration, stochastic/noiseless channel, estimator, and uncertainty. The chain

$$X_t \xrightarrow{\mathcal{R}} \tilde{X}_t \xrightarrow{\mathcal{O}_\lambda} Z_t \xrightarrow{\mathcal{K}_\lambda} \text{Law}(Y_t | Z_t) \xrightarrow{\hat{\theta}_\lambda} \hat{\theta}_t$$

keeps the physical state, ideal sensor signal, recorded data, and estimate distinct.

Boundary 1.1 (Two admissible model modes). A prescribed pair (γ, v) defines a kinematic benchmark. A dynamical prediction requires mass, forces, constraints, drive, dissipation, and initial data. The speed may not be simultaneously prescribed and claimed as the output of an unspecified force law.

2 Track as an adapted framed curve

Let

$$\gamma : [0, L] \rightarrow \mathbb{R}^3, \quad \|\gamma'(s)\| = 1,$$

be C^3 , and set $T = \gamma'$. An adapted frame is

$$R(s) = (e_1, e_2, e_3) \in \text{SO}(3), \quad e_1 = T.$$

It may be a Bishop/rotation-minimizing frame with an added roll profile, or any other declared C^2 adapted frame. Unlike the Frenet frame, this definition remains meaningful at points where $\gamma'' = 0$, provided R is supplied smoothly.

Definition 2.1 (Frame strain). The spatial frame strain $\kappa_R^B(s)$ is defined by

$$R^\top R' = [\kappa_R^B]_\times, \quad [\kappa]_\times u = \kappa \times u. \quad (1)$$

It has dimension L^{-1} .

Curvature magnitude is

$$\kappa(s) = \|T'(s)\|.$$

Torsion is defined through the Frenet frame only where $\kappa > 0$ with sufficient regularity. The supplied roll/frame profile remains the primary apparatus orientation.

3 Time kinematics

Let $s = s(t)$ with $v = \dot{s} > 0$. Then

$$\dot{\gamma} = vT, \quad (2)$$

$$a^I = \ddot{\gamma} = \dot{v}T + v^2T' = v \frac{dv}{ds}T + v^2T'. \quad (3)$$

Since

$$R^\top \dot{R} = vR^\top R' = [v\kappa_R^B]_\times,$$

the vehicle-frame angular velocity is

$$\Omega^B = v\kappa_R^B. \quad (4)$$

Let g^I be gravitational acceleration and let $g_0 > 0$ be a declared reference acceleration. The ideal accelerometer specific force is

$$f^B = R^\top(a^I - g^I). \quad (5)$$

Definition 3.1 (Vector and scalar load).

$$G(s) = \frac{f^B(s)}{g_0}, \quad G(s) = \|G(s)\|. \quad (6)$$

G is a dimensionless body-frame vector; G is a dimensionless scalar. They cannot share one quantity identifier.

Definition 3.2 (Body-frame jerk signal). The derivative of the measured specific-force components is

$$j_f^B = \frac{df^B}{dt} = R^\top(\dot{a}^I - \dot{g}^I) - \Omega^B \times f^B. \quad (7)$$

The last term is required because the sensor axes rotate. The inertial jerk $\dot{a}^I$, the derivative of specific force, and the derivative of body components are different observables.

4 Rail offsets and their domain

Let $d > 0$ be the ideal rail gauge and let $e_2(s)$ be the labeled lateral frame direction. Define

$$\gamma_\pm(s) = \gamma(s) \pm \frac{d}{2}e_2(s). \quad (8)$$

Proposition 4.1 (Sufficient local regularity bound). *If*

$$\frac{d}{2} \sup_{s \in [0, L]} \|e'_2(s)\| < 1, \quad (9)$$

then both offset curves in equation (8) are regular.

Proof.

$$\|\gamma'_\pm\| = \|T \pm (d/2)e'_2\| \geq 1 - (d/2)\|e'_2\| > 0.$$

□

The bound is sufficient, not necessary. It prevents a local offset cusp but not a global self-intersection. To interpret the offsets as two distinct rails, the strip map

$$\Psi(s, u) = \gamma(s) + ue_2(s), \quad |u| \leq d/2,$$

must additionally be an embedding (or be restricted to a declared tubular neighborhood).

The rail lengths are

$$L_\pm = \int_0^L \|\gamma'_\pm(s)\| ds.$$

For a labeled frame define

$$\Delta L_{\text{or}} = L_+ - L_-, \quad \Delta L_{\text{abs}} = |L_+ - L_-|.$$

Proposition 4.2 (Transformation law of the split). *Under a static Euclidean isometry applied to the entire framed track, $L_{\pm}$, ΔL_{or} , and ΔL_{abs} are unchanged. Under the label exchange $+ \leftrightarrow -$,*

$$\Delta L_{\text{or}} \mapsto -\Delta L_{\text{or}}, \quad \Delta L_{\text{abs}} \mapsto \Delta L_{\text{abs}}.$$

Thus the signed split is a covariant of an oriented/labeled rail system, not an unqualified invariant.

5 Rolling count and contact qualification

For constant effective wheel radius $r_w > 0$, pure rolling without slip along rail $\gamma_{\pm}$ gives the ideal revolution count

$$N_{\pm}^{\text{ideal}} = \frac{L_{\pm}}{2\pi r_w}.$$

This formula is dimensionless and parameterization independent. It is not an actual wheel count if slip, flange contact, compliance, wear, variable rolling radius, or loss of contact occurs. Those effects require a contact model and a wheel-angle observation channel.

6 Force and dissipation layer

Let F_d^I be the total dissipative spatial force on a car at a declared application point and v_{rel}^I its velocity relative to the corresponding support or medium. The common sign convention is

$$P_{\text{diss}} = -F_d^I \cdot v_{\text{rel}}^I \geq 0.$$

For a stationary ideal rail constraint, the normal reaction performs no work along an admissible motion. With generalized Lagrangian energy E_L ,

$$\dot{E}_L = P_{\text{drive}} - P_{\text{diss}} + P_{\text{con}} - \partial_t L.$$

For a static autonomous track with ideal normal constraints, $P_{\text{con}} = 0$. A launch system, moving track element, or active brake is not covered by that simplification.

Boundary 6.1 (No force from geometry alone). Curvature determines the kinematic acceleration required by a chosen speed. It does not by itself determine wheel loads, structural stress, or loss of contact. Those require a mass/inertia model and solved constraint reactions.

7 Observation loss

Consider the scalar load observation

$$\mathcal{O}_G : (\gamma, R, v) \mapsto G(s) = \frac{\|a^I(s) - g^I\|}{g_0}.$$

Proposition 7.1 (Scalar load is not a complete framed-track observation). *Fix a centerline γ and speed v . Any two smooth adapted frames R_1, R_2 with the same tangent but distinct roll profiles produce the same scalar load G . Hence $\mathcal{O}_G$ is not injective on framed tracks.*

Proof. Equation (3) depends on γ, v , not on the choice of lateral/up axes. Orthogonality gives

$$\|R_i^T(a^I - g^I)\| = \|a^I - g^I\|$$

for both frames, although their body-component vectors generally differ. $\square$

The full vector G , angular velocity, visual flow, contact signals, and known speed add information but are still protocol outputs, not automatically a complete reconstruction of the track.

8 Transformation audit

Object	Transformation group	Status
Arc length and curvature magnitude	static Euclidean isometries	invariant
Frame components G, Ω^B	declared apparatus-axis change	covariant vectors
Scalar load G	apparatus-axis orthogonal change	invariant
Signed rail split ΔL_{or}	rail-label exchange	sign-covariant
Absolute split ΔL_{abs}	rail-label exchange	invariant
Ideal wheel count	curve reparameterization	invariant under orientation-preserving regular reparameterization
Body jerk components	time-dependent apparatus frame	require the correction in equation (7)

9 Reproducibility and claim boundary

A numerical benchmark must report:

- (i) γ, R, v and their differentiability;
- (ii) sampling and the derivative/smoothing method;
- (iii) the bound equation (9) and global strip validation;
- (iv) gravity, sensor axes, and force/contact status;
- (v) numerical convergence.

The v1.0 synthetic numbers are not retained as engineering results: the layout supplied no clearance, structural, contact, or convergence certificate.

This document does not claim:

- a realizable or safe track;
- a multibody wheel-rail solution;

- structural, fatigue, restraint, or human-tolerance validation;
- recovery of a track from one rider signal;
- novelty of the standard curve and accelerometer formulas.

References

- [1] R. L. Bishop, "There is more than one way to frame a curve," *American Mathematical Monthly* 82 (1975), 246–251. doi:10.2307/2319846.
- [2] M. P. do Carmo, *Differential Geometry of Curves and Surfaces*, Prentice-Hall, 1976.
- [3] B. Brogliato, *Nonsmooth Mechanics: Models, Dynamics and Control*, 3rd ed., Springer, 2016. doi:10.1007/978-3-319-28664-8.